

RAG and Context Engineering
Wikipedia Retrieval Engine - Capstone Design Review

Author: Lakshmi Muthukumar



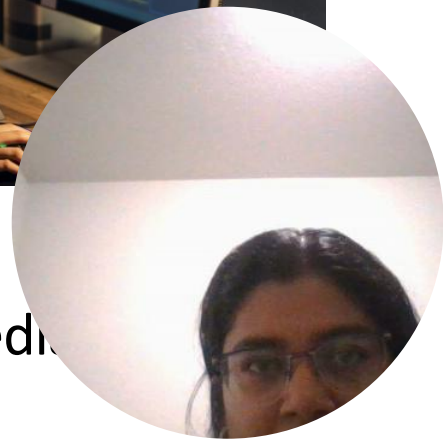
LEADER LENS

WIKIPEDIA RETRIEVAL & EVIDENCE ENGINE









Designing & Evaluating a Retrieval-Augmented LLM System

A retrieval-augmented generation system for grounded, evidence-backed Wikipedi
world leaders.



The Problem: Grounding LLM Answers

LLMs Hallucinate

General-purpose models answer beyond their training data, conflating leaders, dates, and roles.

Historical Facts Are Fragile

Leaders share countries, offices, names, and overlapping political eras—precision is non-negotiable.

Production Requires Proof

Correct-looking answers aren't enough. Retrieval recall, groundedne
latency must all be measurable.



Leader Lens: Corpus, Ground Truth & Intended Audience

What the system knows • What counts as correct • Who it serves

CORPUS — What does Leader Lens know?



65 validated Wikipedia world-leader articles

22 Presidents • 17 Prime Ministers
13 Monarchs • other leaders

- ✓ Precise Fact Retrieval
- ✓ Article-Grounded Answers
- ✓ Exact Evidence Retrieval
- ✓ Cross-Article Synthesis
- ✓ Multi-Hop Reasoning
- ✓ Traceable Answers

GROUND TRUTH — What counts as correct?

EVIDENCE BEFORE INFERENCE

The validated Wikipedia corpus is the source of truth — not the model's memory.

Question
↓
Retrieve corpus evidence
↓
Inspect / expand context
↓
Generate from evidence
↓
Cite supporting passages

Correct • Complete • Grounded • Traceable

If the corpus does not support a claim, the system should not guess.

AUDIENCE — Who is this for?

LEARNERS & EDUCATORS

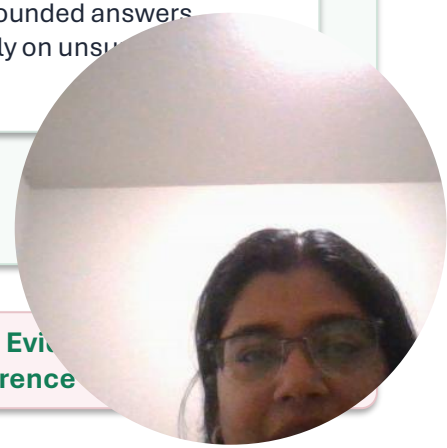
Verify roles, dates, countries, events, and historical context.

RESEARCHERS & ANALYSTS

Compare leaders, offices, political systems, periods, and transitions with traceable evidence.

WRITERS & FACT-CHECKERS

Obtain evidence-grounded answers without relying solely on unsu… model memory.



65 validated articles → increasingly complex retrieval tasks → evidence-grounded answers

Core principle: Evi… inference

From a Simple Lookup to a Multi-Hop Evidence Problem

SIMPLE LOOKUP

“When did Nelson Mandela serve as president?”

Traditional search can usually answer this with one direct fact retrieval.

WHY RAG GETS HARDER

ONE FACT	EVIDENCE CHAIN
Question	Question
↓	↓
Retrieve	Retrieve
↓	↓
Answer	Expand context
	↓
	Connect evidence
	↓
	Verify
	↓
	Answer

MULTI-HOP QUESTION

Using the Wikipedia article on Abraham Lincoln, identify his presidential number and years served; then connect the Gettysburg Address to the Battle of Gettysburg, the Civil War, and Lincoln’s broader vision of unity, equality, and democracy. Conclude with the chronology and supporting evidence.

HOP 1	Lincoln’s presidential number and term
HOP 2	Date and location of the Gettysburg Address
HOP 3	Relationship between the Battle of Gettysburg and the speech
HOP 4	Connection to the Civil War and Lincoln’s political vision
HOP 5	Chronological synthesis with supporting evidence

WHY AGENTIC RETRIEVAL HELPS

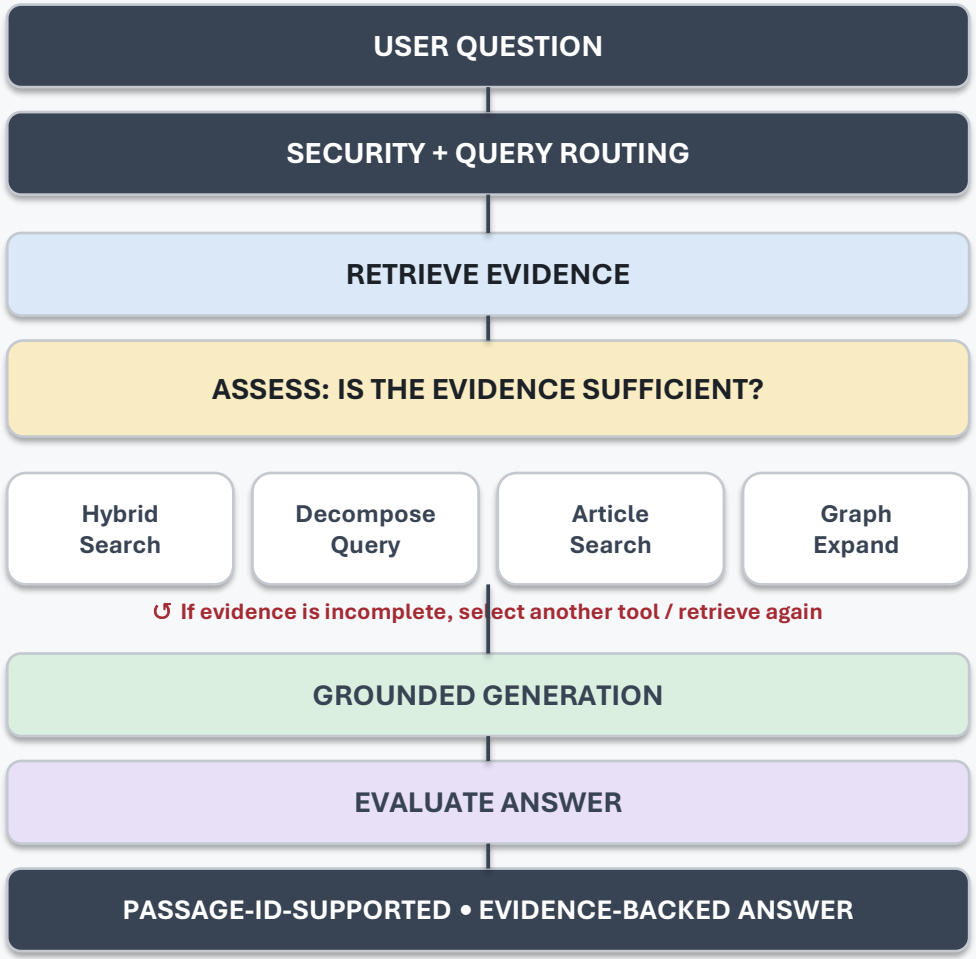
Retrieve → Expand → Connect → Verify → Synth



The Solution: Leader Lens System Architecture

Tool-Using Agentic RAG with shared retrieval infrastructure and three control strategies

1. HIGH-LEVEL ARCHITECTURE



2. WORKFLOW COMPONENTS

01 — Security & routing

Protects the request and directs it to the appropriate retrieval path.

02 — Hybrid retrieval

Combines BM25 lexical search with vector semantic search.

03 — Agentic evidence review

Checks whether retrieved evidence covers every part of the question.

04 — Tool selection

Chooses hybrid_search, decompose_query, article_search, or graph_expand based on the evidence need.

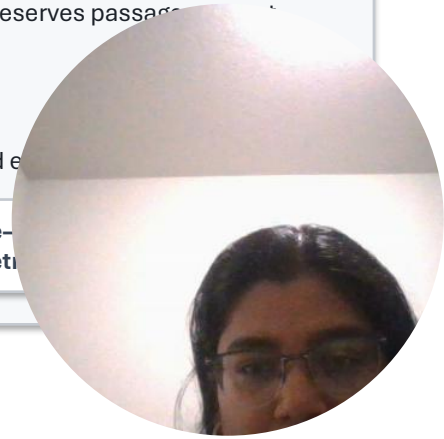
05 — Grounded generation

Builds the response only from retrieved passages and preserves passag

06 — Evaluation

Checks answer quality, grounding, security, latency, and e

CONTROL PATHS Fixed = one pass • Agentic = retrieve–
Tool-Using = agent selects ret



How does the system work?

One question is tested through three retrieval strategies, then judged against the same ground truth.

QUESTION
Wikipedia query

01

FIXED

One hybrid search

BM25 + vector
→ top-k evidence

Fast baseline

02

AGENTIC

Retrieve → inspect

Refine query only
when evidence is weak

Adaptive retrieval

03

TOOL-USING

LLM selects tools

Hybrid search • article search • graph expansion

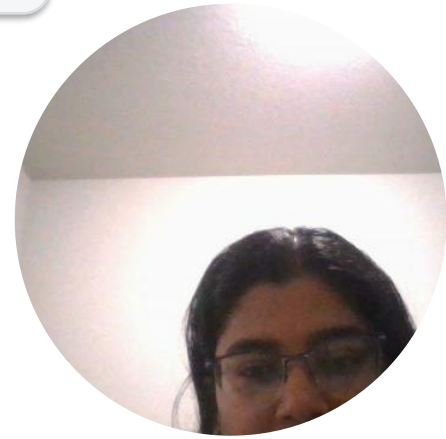
Most retrieval control

EVIDENCE-ONLY ANSWER

Generate from retrieved passages and cite passage IDs.

LLM JUDGE

Compare answer to XLSX grading notes → score quality + PASS / PARTIAL / FAIL



Checkpoint 5.1 • Hybrid retrieval uses BM25 + vector search • Agent is bounded to MAX_STEPS = 3

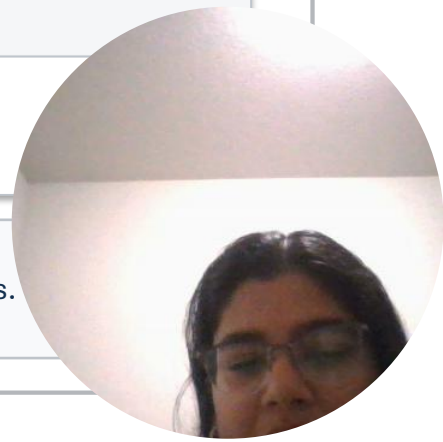
The agents retrieve smarter — but at a cost!

Leader Lens | **compares three retrieval strategies.**

Same corpus and retriever; increasing autonomy, retrieval depth, and reasoning.

Characteristic	Fixed Retrieval	Agentic Retrieval	Tool-Using Agent
How it operates	One hybrid pass → grounded answer	Retrieves → reviews evidence → refines queries when needed	Chooses among decomposition, article search, graph expansion, and hybrid search
Best observed strength	Fast and predictable when first-pass evidence is sufficient	Recovered broader evidence when initial retrieval was incomplete	Targeted retrieval and cross-article evidence gathering
Observed limitation	Can miss facts outside the initial top passages	Can continue until the round limit, increasing latency	Tool routing can repeat actions or hit MAX_STEPS = 3
Representative results	Nelson Mandela: 4 passages / 4.65 s Jobs: 4 / 4.50 s Cross-doc: 4 / 8.27 s	Nelson Mandela: 11 / 17.71 s Jobs: 8 / 15.63 s Cross-doc: 19 / 26.69 s	Nelson Madela:: 9 / 10.93 s Jobs: 6 / 7.13 s Cross-doc: 15 / 22.25 s
Average latency (3 representative tests)	**5.81 s**	**20.01 s**	**13.44 s**

RESULT: Agentic retrieval improved evidence coverage on harder questions, but added latency and step/round-limit trade-offs.



System design — what did the retrieval strategy change?

Evidence-first answers from a local Wikipedia corpus, with increasing retrieval control from Fixed → Agentic → Tool-Using.

Same questions + same grading notes; only the retrieval strategy changes.

81.3
mean GT score
best of the 3 workflows

40%
PASS rate
vs 20% Fixed

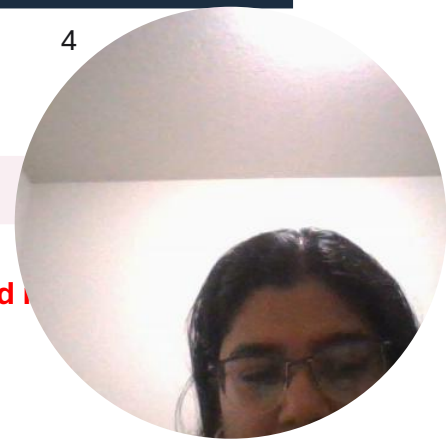
0%
FAIL rate
vs 40% Fixed

5.29s
mean latency
10 passages / question

Workflow comparison

Workflow	Mean score	Pass	Fail	Latency	Passages
Fixed	67.5	20%	40%	2.09s	4
Agentic	76.2	20%	20%	4.77s	
Tool-Using	**81.3**	**40%**	**0%**	**5.29s**	

Takeaway: Tool-Using retrieval improved quality and eliminated failures in this 5-question run, but required more evidence and l



Evaluation set: 5 questions • Scores are from the standalone GPT-5.4 judge against the XLSX grading notes

How are the answers evaluated?

Generation quality is judged against the ground-truth grading notes; the overall score is deterministic.

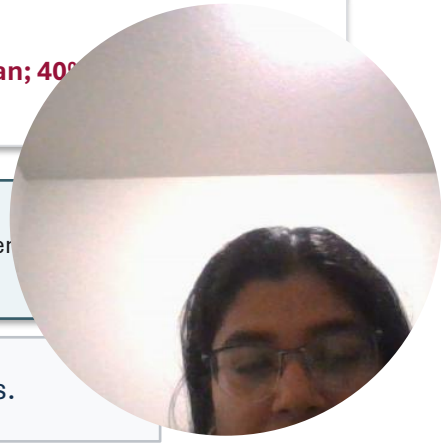
Metric	What it measures	How it is scored	Checkpoint result
Correctness	Is the final answer factually right relative to the grading notes?	LLM judge: 0–4	Tool-Using mean: 2.6 / 4
Completeness	Does the answer cover every requested component?	LLM judge: 0–4	Tool-Using mean: 2.6 / 4
Groundedness	Are claims visibly tied to retrieved passage IDs and source-aligned?	LLM judge: 0–4	Tool-Using mean: 3.8 / 4
Source compliance	Did the answer use the required Wikipedia article(s)?	LLM judge: 0–4	Tool-Using mean: 4.0 / 4
Overall score	Single comparable quality score across workflows	(sum of four scores / 16) × 100	**81.26 mean; 40**

PASS ≥ 85 • PARTIAL 60–84.9 • FAIL < 60

Thresholds are applied deterministically after judging.

Important limitation

Standalone groundedness checks citation coverage/source alignmer
text.



RESULT: Agentic retrieval improved evidence coverage on harder questions, but added latency and step/round-limit trade-offs.

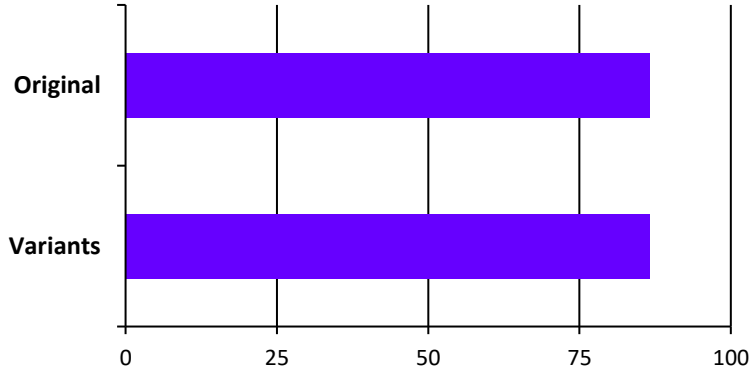
Retrieval & Evaluation: Where Leader Lens Succeeds—and Where It Breaks

15	60	GPT-5.4	PASS / FAIL
canonical questions	total tests	independent judge	evaluation

Representative evaluation boundary

Single-document factual retrieval	PASS
Multi-fact retrieval	PASS
Leadership-role retrieval	PASS
Multi-document synthesis	PASS
Cross-document comparison	PASS
Exact quotation retrieval	FAIL

Paraphrase robustness



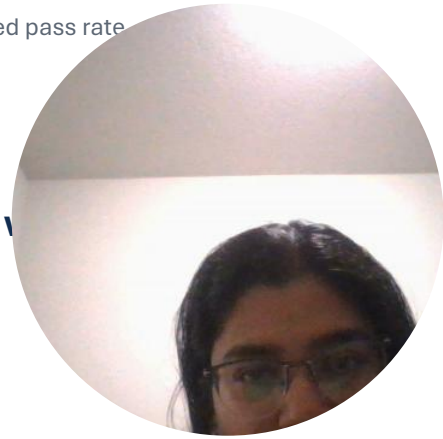
68.7% original → 70.5% variants

Paraphrasing did not reduce the measured pass rate

Reliable across focused and multi-document retrieval. Exact quotation recovery remains the clearest v

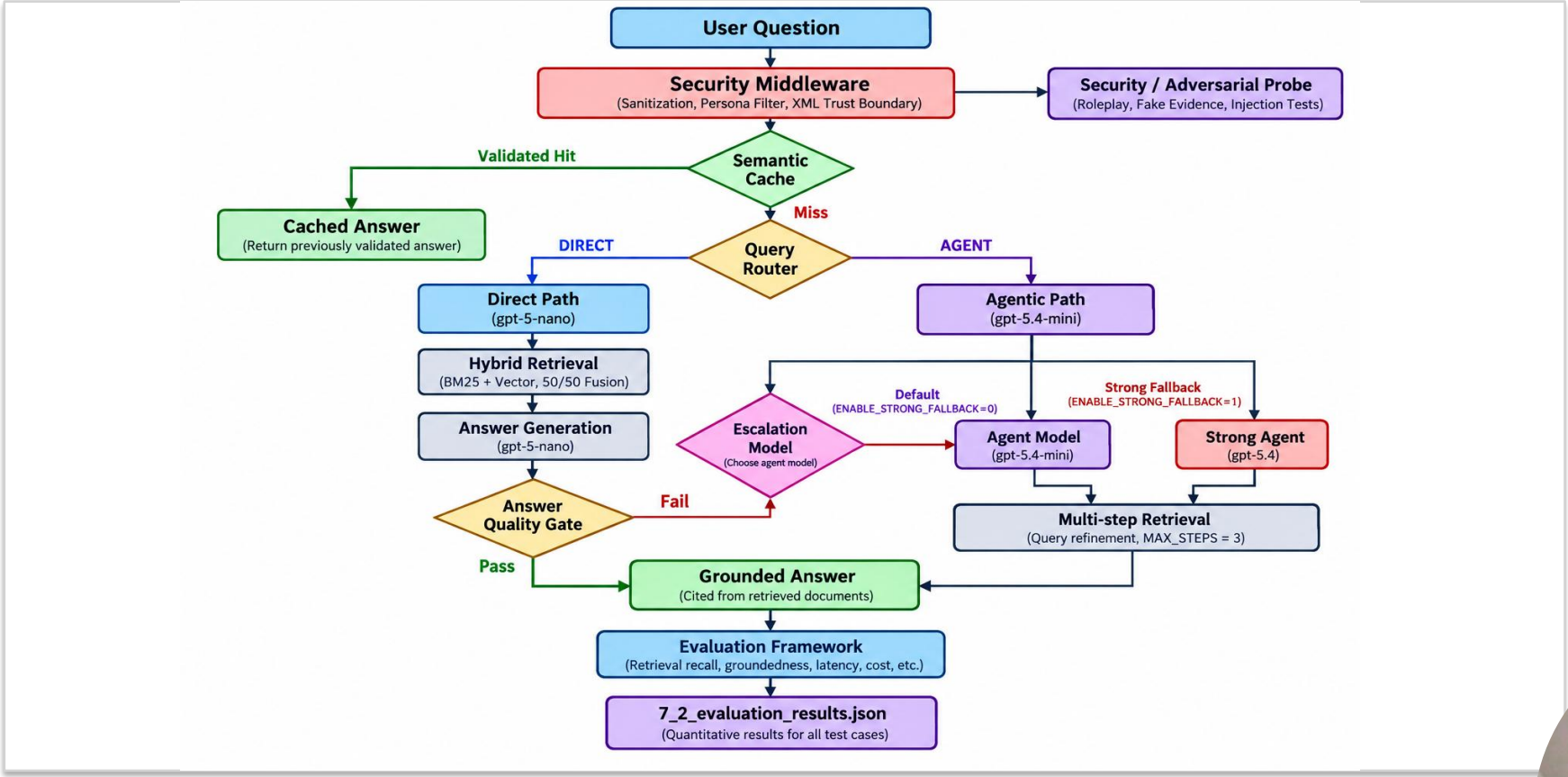
Multi-document synthesis and cross-document comparison pass; exact-passage retrieval is the remaining representative failure.

13/15 canonical questions passed • 39/45 paraphrase variants passed • Independent GPT-5.4 judge

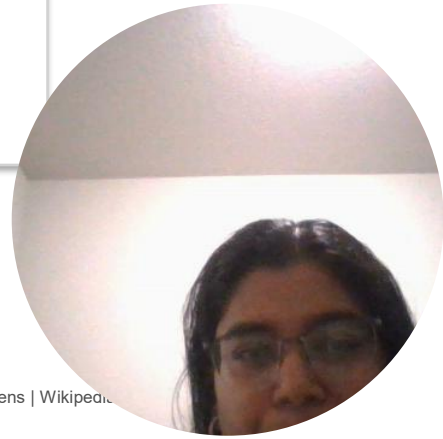


Engineering Robust Retrieval-Augmented Systems

LEADER LENS PRODUCTION ARCHITECTURE



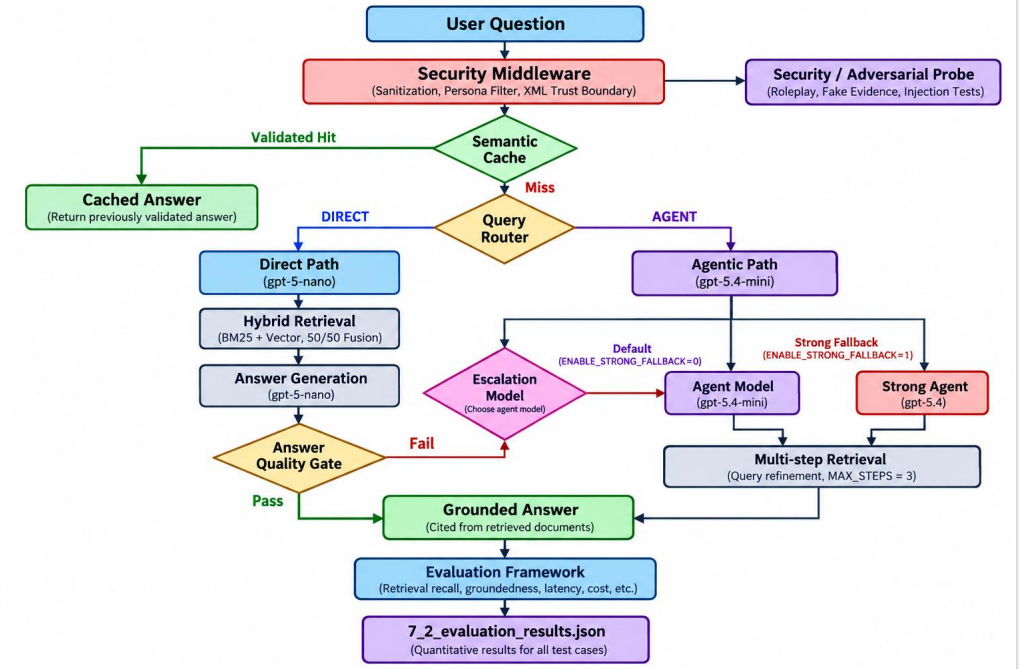
Protect → Route → Retrieve → Validate → Ground



Leader Lens | Wikipedi

Engineering Robust Retrieval-Augmented Systems

Leader Lens production architecture



How the request moves through the workflow

PROTECT & REUSE | ROUTE & RETRIEVE | CHECK & ESCALATE | GROUND & MEASURE

1 PROTECT & REUSE

Security middleware sanitizes the question and enforces the trust boundary. The semantic cache then checks for a validated high-similarity answer; a safe hit returns immediately.

2 ROUTE & RETRIEVE

On a cache miss, the router selects DIRECT or AGENT. The direct path uses gpt-5-nano + hybrid retrieval; complex comparison and multi-hop questions use gpt-5.4-mini with iterative retrieval.

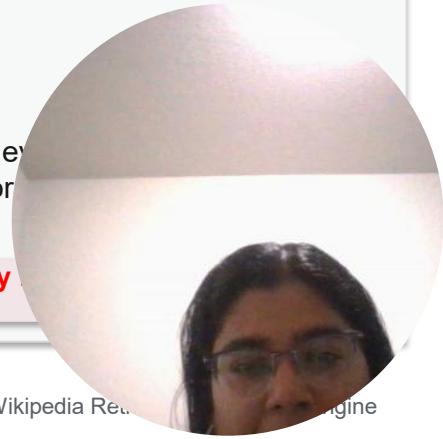
3 CHECK & ESCALATE

The direct-path quality gate accepts supported answers and escalates weak ones to the agent. Agentic retrieval is bounded at MAX_STEPS = 3, with optional gpt-5.4 fallback for harder cases.

4 GROUND & MEASURE

The final answer is generated from retrieved Wikipedia evidence. The ev
records routing, retrieval recall, groundedness, latency, cache behavior
cost.

**Design principle: spend model capability selectively while keeping every
grounded and measurable.**

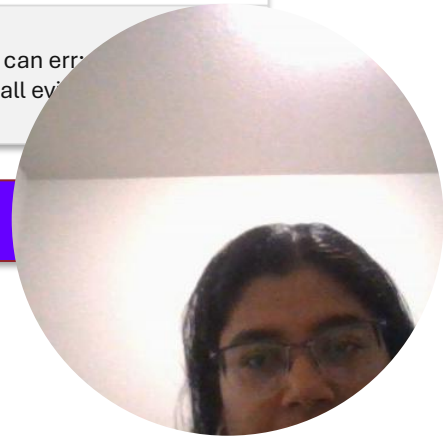


Engineering Decisions

Design decision	Alternative considered	Why we chose it	Trade-off
50/50 Hybrid Retrieval BM25 + Vector	BM25-only or vector-only	Leader queries need exact names/dates AND semantic historical context. Fusion combines lexical precision with semantic recall.	More retrieval complexity; fixed weights may not be optimal for every query.
Query Router: Direct vs. Agentic	Send every query through the full agent	Simple one-hop questions use the cheap direct path; comparisons and multi-hop questions get iterative retrieval only when needed.	Router can misclassify; requires a quality gate and escalation.
Quality-gated Escalation	Accept every direct-path answer	A low-cost direct answer is accepted only when evidence supports it; otherwise it escalates to GPT-5.4-mini.	One extra quality-check call adds some latency.
Validated Semantic Cache	Exact-match cache or no cache	High-similarity paraphrases can reuse answers; corpus/config signatures prevent reuse after the system changes.	High threshold lowers hit rate; lower thresholds risk incorrect reuse.
Security Middleware + XML Trust Boundary	Prompt instructions alone	Sanitize before retrieval and separate untrusted questions from trusted Wikipedia evidence.	Extra preprocessing/filtering; sanitizer can alter legitimate input.
Bounded Agent + Tiered Models	Strongest model for every request	Nano models handle lightweight decisions; GPT-5.4-mini handles reasoning; MAX_STEPS=3 bounds cost and latency.	Small models can err; stop before all ev

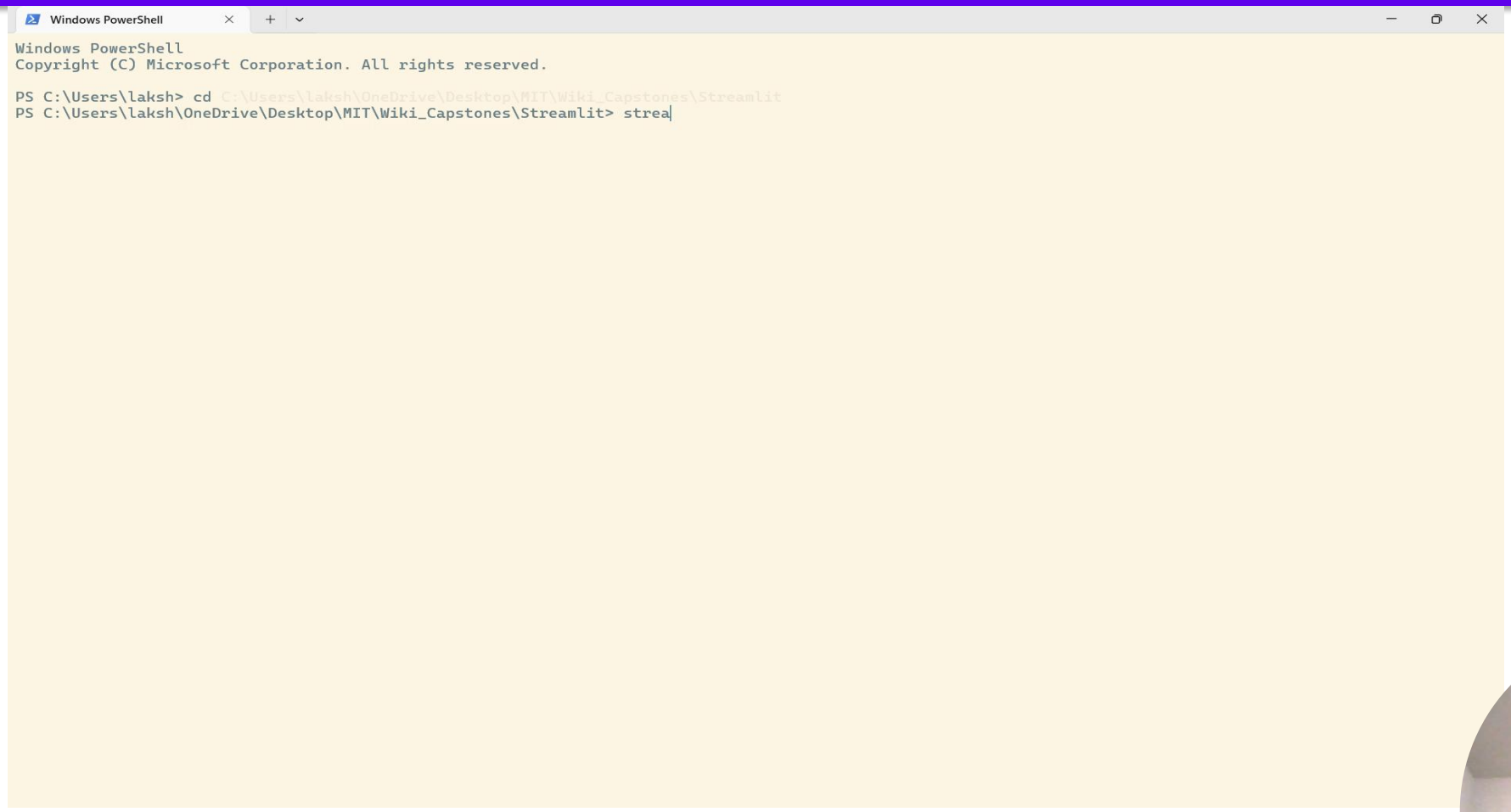
Why this final approach

- **Evidence first:** hybrid retrieval and grounded generation prioritize Wikipedia evidence over unsupported model knowledge.
- **Spend intelligence selectively:** cache → direct → quality gate → agent → optional strong fallback.
- **Production balance:** optimize answer quality, security, latency, token cost, and bounded autonomy together—not one metric in isolation.

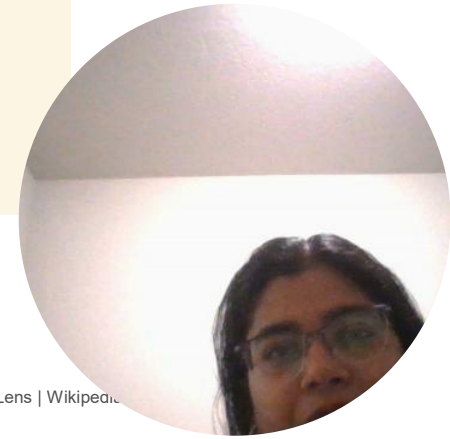


Engineering Robust Retrieval-Augmented Systems

LEADER LENS — LIVE SYSTEM DEMONSTRATION



Protect → Route → Retrieve → Validate → Ground



Leader Lens | Wikipedi

Validation Trade-offs — Efficiency Gains and Remaining Cost

A controlled cache demonstration shows where production optimization delivers immediate value.

Scenario	What changed	Latency	Tokens	Cost implication
Cache seed	Full retrieval + generation	21.85 s	4,546	Baseline generation cost
Paraphrase hit	Validated semantic-cache reuse	**0.066 s**	**0**	**Generation cost avoided**

99.7%

latency reduction

4,546 → 0

generation tokens

21.85 → 0.066 s

response latency

Dollar cost note

The run did not calculate USD cost because model-price environment variables were unset. Report token avoidance—not an invented dollar saving.

TRADE-OFF

Validated cache reuse can nearly eliminate latency and generation work for repeated intent, but the cache must remain conservative enough to avoid reusing an answer for a materially different question.

Measured evidence: cache MISS = 4,546 tokens / 21,853 ms; paraphrase = semantic-cache-hit / 0 tokens / 66.4 ms.



Production Readiness — Security, Agent State & Measured Results

Leader Lens hardens the retrieval loop, bounds agent behavior, and measures what still fails.

Production-ready agent workflow

1 PROTECT INPUT

Sanitize + XML escape + persona/injection filter

2 ROUTE & REUSE

Semantic cache → DIRECT or AGENT

3 TRACK AGENT STATE

pending queries • executed queries • collected passage IDs

4 RETRIEVE & REASON

Hybrid + multi-step evidence gathering ; refine until enough evidence

5 BOUND & GROUND

MAX_STEPS = 3 → evidence-only answer → quality gate

AGENT STATE Original question, evidence collected, queries already executed, and planner decision. -- > done / new_queries / reasoning.

What the production evaluation actually showed

90%	100%	87.5%	59.2%
retrieval recall	safeguard pass	router accuracy	mean groundedness

SECURITY + CACHE DEMONSTRATIONS

✓ Probe rejected pirate roleplay and fabricated Trudeau claims ✓ Cache paraphrase hit: 0 tokens / 66.4 ms

All three adversarial cases used the hardened path; groundedness ≈ 0.83 / 0.86 / 0.90.

PERFORMANCE BASELINE

P50 10.35 s P95 34.93 s

9,928 tokens / question

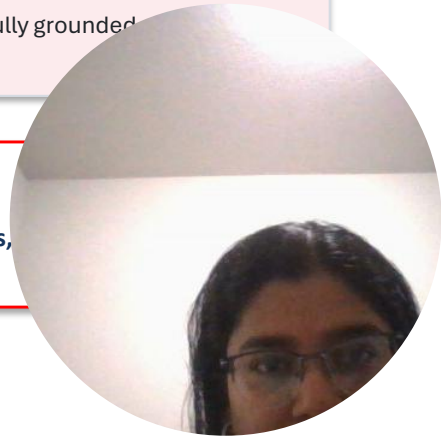
148,913 total tokens

READINESS GAPS

Router misses multi-hop cases
Grounding trails retrieval recall
Some multi-document retrieval is incomplete
Perfect retrieval ≠ fully grounded

PRODUCTION READINESS VERDICT

Security and cache controls are promising; routing, evidence completeness, synthesis are the next engineering priorities.



Evidence base: Capstone_checkpoint_7_1.py

Observed Failures → Evidence → Next Engineering Steps

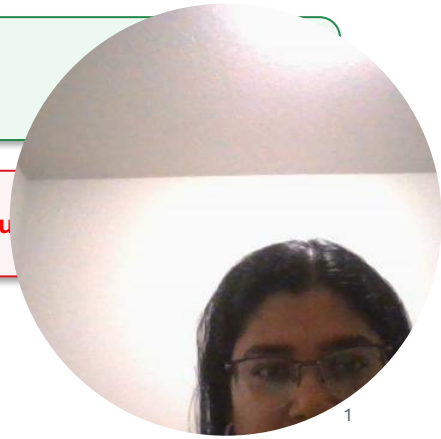
Each engineering change maps directly to a failure reproduced in the current evaluation.

Observed failure	Evidence from current run	Next engineering step
Routing mismatch	**Router accuracy 73.3%; Trudeau, Victoria and Lincoln cases expected AGENT but routed DIRECT.**	Add deterministic complexity signals for multi-part/comparison queries; calibrate the router on labeled cases.
Missing required sources	**Mandela–Suharto and four-leader comparisons reached retrieval recall = 0.50.**	Track required articles as pending retrieval goals; retry article_search before completion.
Retrieved ≠ grounded	**Mean recall 90.0% vs groundedness 59.2%; Washington/Lincoln/Obama: recall 1.0, groundedness ≈ 0.36.**	Add claim-to-evidence coverage checks and require support for every requested fact.
Complex-agent cost	**Washington/Lincoln/Obama consumed 33,392 tokens.**	Cap retained evidence, deduplicate passages, summarize intermediate state, and stop when evidence requirements are satisfied.

WHAT ALREADY WORKED

The Trudeau adversarial probe rejected the pirate roleplay and fabricated “18th PM / Conservative Party” evidence. Preserve this as a regression test.

FIRST PRIORITY: do not complete while a required source or requested sub-question remains unresolved. • Design principle: retrieval presence ≠ grou



Limitations and Future Improvements

Remaining limitations

- **Semantic cache needs broader validation:** the controlled Trudeau paraphrase test succeeded (21.85 s → 0.066 s; 4,546 → 0 generation tokens), but the 15-case evaluation itself contained no cache hits.
- **Grounding remains incomplete:** mean groundedness was 59.2% even though mean retrieval recall remained 90.0%.
- **Complex queries are expensive:** the Washington/Lincoln/Obama comparison consumed 33,392 tokens.
- **Evaluation scale is limited:** 15 cases are useful for the capstone, but do not represent production-scale diversity, concurrency, or adversarial complexity.

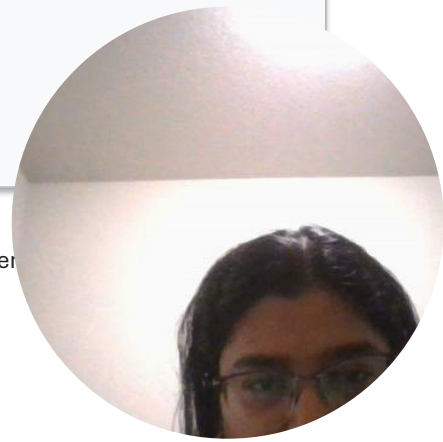
Known risks

- **Routing errors:** 73.3% router accuracy means some multi-part questions were sent DIRECT when AGENT was expected.
- **Missing-source risk:** Mandela–Suharto and the four-leader comparison retrieved only 50% of the expected source coverage.
- **Evidence ≠ grounded answer:** Washington/Lincoln/Obama retrieved all expected articles but achieved groundedness of only ≈0.36.
- **Agentic cost growth:** repeated planning, retrieval, and synthesis can sharply increase token use and latency.

Future enhancements

- **Improve routing:** add deterministic complexity signals for comparisons, multi-part requests, and multiple named entities before LLM routing.
- **Track retrieval obligations:** keep required articles/sub-questions pending and retry targeted article_search before allowing completion.
- **Strengthen grounding:** add claim-to-evidence coverage checks and require support for every requested fact before answer release.
- **Control agent cost:** deduplicate retained passages, summarize intermediate state, cap evidence growth, and stop when all evidence requirements are satisfied.

The system is more secure and measurable, but the next engineering priority is evidence completeness: retrieve every required source, prove claim-level coverage, ther



Lessons Learned — What Building Leader Lens Taught Me

Production RAG is not about retrieving more—it is about retrieving the right evidence, knowing when to stop, and proving every claim.

01 Retrieval ≠ Grounding

Finding the right source does not guarantee a supported answer.

90% retrieval recall vs. 59.2% groundedness

02 More Agentic ≠ Always Better

Agentic retrieval helps on multi-hop questions; simple lookups benefit from the faster DIRECT path.

Use autonomy selectively

03 Evidence Completeness Must Be Explicit

Track required articles and sub-questions; do not finish while evidence obligations remain unresolved.

Retrieve → verify coverage → answer

04 Optimization Needs Guardrails

Caching, routing, smaller models, and bounded agents save work only when paired with validation and escalation.

Optimize without weakening trust

05 Evaluation Drives Architecture

Measured failures in routing, grounding, source coverage, latency, and cost define the next engineering iteration.

Failures become engineering requirements

BIGGEST LESSON: Evidence before inference — retrieve every required source, prove claim-level support, then answer.



Questions & Discussion

Thank you!

Leader Lens was designed as a source-grounded, tool-using RAG workflow—not a general-purpose chatbot.

Be prepared to discuss:

Design rationale

Why hybrid retrieval, agentic control, routing, and bounded tool use?

Evaluation evidence

What do the retrieval, grounding, robustness, security, and latency results show?

Engineering trade-offs

Where do answer quality, autonomy, security, latency, and cost compete?

Core question: How do we make increasingly capable retrieval systems remain grounded, measurable, and production-ready?

